OLIVIA W LANG

EDUCATION

Ph.D. **Cornell University**, Ithaca, NY **2025**
Dr. B. Franklin Pugh (Computational Biology)
The development of quality control and modular analysis tools that support reproducible genomics

B.A. **Colby College**, Waterville, ME **2015**
Interdisciplinary Computation - Biology
Minors: Mathematics, Chinese

RESEARCH STATEMENT

I am a bioinformatician who has collaborated closely with biochemists in the Pugh Lab and the Cornell Epigenomics Core Facility to build custom tools and analysis workflows for emerging assays such as Benzonase ChIP-exo, designed to exploit the base-pair resolution protein binding profiles their chromatin context, including rotational phasing of nearby nucleosomes. Working in a lab with assay-development expertise has taught me to recognize experimental design choices whose biases often go unnoticed on the computational side so that they can be leveraged or accounted for in analysis or the contruction of training datasets.

I am seeking a role where I can apply this interdisciplinary background to help researchers design rigorous experiments, analyses, and models.

TECHNICAL STRENGTHS

Data types	ChIP-exo, ChIP-seq, CUT&Tag, CUT&RUN, RNA-seq, CoPRO, CAGE, Micro-C, shotgun metagenomics, single-cell multi-omics (AVITI24 Teton)
Programming Languages	Java, Python, shell, R, Perl, JavaScript, MATLAB
Bioinformatics Tools	Aligners, Bedtools, Demultiplexers, Samtools, USCS binaries, Genome assemblers
Python Packages	Pysam, Pandas, SciPy, Matplotlib, Biopython, PyTorch
Data libraries & APIs	SRA tools, ENCODE, 4D Nucleome
Package Managers	Anaconda, Docker, Singularity
Workflow Managers	Galaxy, Snakemake
Computing resources	HPC-Slurm/PBS (Pittsburgh Supercomputing, Penn State ROAR Collab, Cornell BioHPC), VMs (Jetstream2), HTC (OSG)

RESEARCH EXPERIENCE

Dr. William K.M. Lai & Dr. Amy M. Lyndaker, Cornell Epigenomics Core Facility 2025 - Present
Postdoctoral Researcher *Ithaca, NY*

Dr. B. Franklin Pugh, Cornell University 2019 - 2025
Graduate Research Assistant *Ithaca, NY*

Dr. Ilana Brito, Cornell University 2018 - 2019
Graduate Reserach Assistant *Ithaca, NY*

Dr. J. Michael Cherry, Stanford University 2015 - 2017
Associate Biocuration Scientist *Palo Alto, CA*

PUBLICATIONS

Lang OW, Gardner A, Smugereski CM, Lyndaker AM, Pugh BF, Lai WKM. (submitted) *Multi-modal in situ profiling of human cells using the AVITI24 platform characterizes single-cell responses to acute stress.* bioRxiv. doi:10.64898/2026.10.28.762240

Lang OW, Beer B, Zhang D, LeSon C, Pugh BF, Lai WKM. (in-review) *ScriptManager: a platform for scalable and reproducible high-resolution analysis of genomics datasets.* bioRxiv. doi:10.64898/2026.06.14.732163

Chen H, **Lang OW**, Lai WKM, Pugh BF. (in-review) *Fine architecture and mechanism of PIC assembly at natural genomic promoters.*

Cha J, **Lang OW**, Beer B, Pugh BF, Lai WKM. (in-review) *GenoPlotter: interactive composite plotting of quantitative data at genomic loci.*

Chen H, Krebs JE, Lyndaker AM, Smugereski CM, Melini D, **Lang OW**, Gardner AM, Lai WKM, Pugh BF. (in-review) *ChIP-exo v6 protocol for high resolution genome-wide mapping of protein-DNA interactions in context with nucleosomes.*

Chen H, Krebs JE, **Lang OW**, Hu J, Melini D, Hyle J, Li C, Lai WKM, Pugh BF. (2026) *Genome-wide rotational and translational phasing of nucleosomes with human transcription factors.* Mol Cell. doi:10.1016/j.molcel.2026.06.019 (PMID: 42392072)

Gafur J, **Lang O**, Lai WKM. (2025) *Adversarial attack of sequence-free enhancer prediction identifies chromatin architecture.* Bioinformatics. doi:10.1093/bioinformatics/btaf371 (PMID: 40581823)

Lang O, Srivastava D, Pugh BF, Lai WKM. (2023) *GenoPipe: identifying the genotype of origin within (epi)genomic datasets.* Nucleic Acids Res. doi:10.1101/gad.350026.122 (PMID: 36302553)

Mittal C, **Lang O**, Lai WKM, Pugh BF. (2022) *An integrated SAGA and TFIID PIC assembly pathway selective for poised and induced promoters.* Genes Dev. doi:10.1007/978-1-4939-7737-62 (PMID: 36302553)

Lang OW, Pugh BF, Lai WKM. (2022) *ScriptManager: an interactive platform for reducing barriers to genomic analysis.* PEARC'22 2022 Jul. doi:10.1145/3491418.3535161

Lang OW, Nash RS, Hellerstedt ST, Engel SR, SGD Project. (2018) *An Introduction to the Saccharomyces Genome Database (SGD).* Methods Mol Biol. doi:10.1007/978-1-4939-7737-6_2 (PMID: 29761454)

MacPherson KA, Starr B, Wong ED, Dalusag KS, Hellerstedt ST, **Lang OW**, Nash RS, Skrzypek MS, Engel SR, Cherry JM. (2017) *Outreach and online training services at the Saccharomyces Genome Database.* Database. doi:10.1093/database/bax002 (PMID: 28365719)

PRESENTATIONS - SPEAKER

CVG Symposium (2026), Ithaca, NY — 2026 May 15
A spatial and single-cell multi-omic approach to understanding peroxide stress response

Computational Biology Student Seminar, Ithaca, NY — 2023 December 1
Scalable infrastructure design for rigorous and reproducible epigenomic analysis

Computational Biology Field New Student Orientation, Ithaca, NYMA — 2022 August 16
Investigating the mechanisms of transcription regulation

Practice and Experience in Research Computing (PEARC'22), Boston, MA — 2022 Jul 10
ScriptManager: An interactive platform for reducing barriers to genomic analysis

Computational Biology Student Seminar, Ithaca, NY — 2021 Nov 5
GenoPipe: post-sequencing validation of experiment metadata

Computational Biology Student Seminar, Ithaca, NY 2021 April 2
Applying a method for quality control of genetic backgrounds to the Yeast Knockout Collection

OSG User School Lightning Talk, 2020 Jul 24
Bioinformatic Tool Development for Sequencing-based Assays

Computational Biology Student Seminar, Ithaca, NY 2020 May 5
GenoPipe: a QC tool for sample strain background

PRESENTATIONS - POSTERS

Great Lakes Bioinformatics (GLBIO'25), Minneapolis, MN 2025 May 12-15
A spatial and single-cell multi-omic approach to understanding the DNA damage response

Molecular Biology and Genetics Retreat, Ithaca, NY 2023 August 9
Interpreting transcription start site patterns in yeast

PSU 39th Summer Symposium: Chromatin and Regulation of Gene Expression, State College, PA 2023 August 6-9
Interpreting transcription start site patterns in yeast

Northeast Regional Yeast Meeting (NERY'23), Ithaca, NY 2023 July 27-28
Interpreting transcription start site patterns in yeast

Great Lakes Bioinformatics (GLBIO'23), Montreal, QC 2023 May 15-19
GenoPipe: identifying the genotype of origin within (epi)genomic datasets

The Allied Genetics Conference (TAGC), Orlando, FL 2016 July 13-17
The Saccharomyces Genome Database Variant Viewer tool

TEACHING & MENTORSHIP

Graduate Student Mentor Fall 2024 - present
CVG Mentoring Program *Ithaca, NY*

Graduate Student Mentor Fall 2021 - present
Pugh Lab computational undergraduates *Ithaca, NY*

Quantitative Genetics & Genomics Teaching Assistant Spring 2019
Dr. Jason Mezey *Ithaca, NY*

FUNDING & AWARDS

BIO240207: "Screening for key drivers of gene regulation in cell-type specific contexts using machine learning approaches" 2025 November
ACCESS Grant *$18,731.79*

Travel Fellowship for GLBIO'25 2025 May 12-15
Great Lakes Bioinformatics (GLBIO'25)

Research Software Engineering Training Workshop 2023 July 10-14
INTERSECT '23

Travel Fellowship for GLBIO'23 2023 May 15-19
Great Lakes Bioinformatics (GLBIO'23)

Best Paper - Applications and Software Short Paper Track 2023 May 15-19
Practice and Experience in Research Computing (PEARC'22)

Student Program - fellowship to cover travel and special workshops 2023 May 15-19
Practice and Experience in Research Computing (PEARC'22)

ACADEMIC & COMMUNITY SERVICE

Epigenomics Research Group Member 2026-present
Association of Biomolecular Resource Facilities (ABRF)

Training Executive Committee Member 2024-present
Center for Vertebrate Genomics at Cornell *Ithaca, NY*

Youth Workshop Volunteer 2022 April 7
Expanding Your Horizons *Ithaca, NY*

Graduate Student Volunteer Reviewer Spring 2022
Cornell Undergraduate Research Journal *Ithaca, NY*

Graduate Student Honors Thesis Reviewer 2022-2026
Cornell Computational Biology Honors Thesis Program *Ithaca, NY*

Social Chair 2021-2022
Cornell Graduate Women in Science *Ithaca, NY*

Organization Officer; President 2018-2022
Cornell Computational Biology Graduate Student Association *Ithaca, NY*

Organization Officer; Founding member 2019-2023
Touch Rugby at Cornell *Ithaca, NY*

Assistant Coach 2018-2020
Cornell Men's Rugby Team *Ithaca, NY*

Organization Officer 2018-2019
Cornell Grads Vote Student Organization *Ithaca, NY*